

Jiahui Huang

j346huan@uwaterloo.ca | j346huan.github.io | +1 519-277-8529

Education

PhD in Pure Mathematics University of Waterloo, Canada Supervisor: Matthew Satriano	Sept 2023 - April 2027 (expected)
Master in Mathematics ETH Zürich, Switzerland Supervisors: Arkadij Bojko, Rahul Pandharipande Thesis: <i>Equivariant Segre and Verlinde invariants for Quot schemes.</i>	Sept 2021 - April 2023
Bachelor in Pure Mathematics University of Waterloo, Canada Supervisors: Matthew Satriano, David McKinnon	Sept 2017 - June 2021

Publications and Preprints

<i>The dimension threshold for Batyrev's non-negativity conjecture on stringy Hodge numbers.</i> Jiahui Huang, Matthew Satriano. <i>Arxiv preprint.</i>	2026
<i>A cohomological interpretation for stringy Hodge numbers.</i> Jiahui Huang, Matthew Satriano, Jeremy Usatine. <i>Arxiv preprint.</i>	2026
<i>Contact loci of semihomogeneous singularities.</i> Eduardo de Lorenzo Poza, Jiahui Huang. <i>Arxiv preprint.</i>	2025
<i>Equivariant Segre and Verlinde invariants for Quot schemes.</i> Arkadij Bojko, Jiahui Huang. <i>Moduli</i> , 2 e14.	2025
<i>What fraction of an S_n-orbit can lie on a hyperplane?</i> Jiahui Huang, David McKinnon, and Matthew Satriano. <i>Linear Algebra and Its Applications</i> , 613C pp. 1-23.	2021

Honours and Awards

PMATH Outstanding Teaching Assistant Award	2025
UW Graduate Scholarship \$2385	2025
MATH Senate Graduate Scholarship \$2000	2024
Pure Math Graduate Entrance Scholarship \$5229	2024
Women in Math Mentorship Award \$2000 <i>This award recognizes the commitment invested in helping undergraduate students who are women or gender minorities to develop insight and skills in Mathematics, valued at \$1000 each.</i>	2024, 2025
Undergraduate Research Assistanceship <i>This award provides research experience that nurtures an interest in research careers and mathematics experience with financial support during a full-time, on-campus position, totaling \$12000.</i>	2019, 2020
President's Scholarship \$2000	2018
J.D. Kalbfleisch Scholarship for International Students \$15750 <i>Several awards, valued at \$1750 each, are provided across undergraduate study.</i>	2017-2021

Teaching

Calculus for Sciences (lectures, notes, office hours)	2025
---	------

Teaching Assistantships

Advanced Linear Algebra 2 (grading, guest lectures, office hours)	2026
Representation Theory (grading, guest lectures)	

Linear Algebra (tutoring)	
Algebraic Geometry (grading, office hour, supplement notes)	2025
Algebraic Number Theory (grading, office hour)	
Linear Algebra 2 (grading, office hour, guest lecture)	
Advanced Linear Algebra 3 (grading)	2024
Linear Algebra 2 (grading, guest lecture)	
Algebraic Number Theory (grading, office hour)	
Algebraic Topology (grading)	
Advanced Linear Algebra 2 (grading)	
Linear Algebra (tutoring)	2023
Advanced Linear Algebra 1 (grading)	
Linear Algebra (grading)	2020
Calculus and Linear Algebra (tutoring)	2019
Calculus and Linear Algebra (tutoring)	2018

Others

Volunteer for Graduate Student Association.	2024-2026
Volunteer for CEMC contest.	2025-2026

Talks

<i>Contact Loci of Semihomogeneous Singularities.</i>	2025
Florida State University	
<i>Arc-Floer conjecture for isolated homogeneous singularities.</i>	2024
Ontario Graduate Mathematics Conference, Waterloo	
<i>Arc-Floer conjecture for double point singularities.</i>	2024
Canadian Mathematical Society Summer Meeting, Saskatoon.	
<i>Equivariant Segre and Verlinde invariants for Quot schemes.</i>	2023
Algebraic geometry and moduli seminar, ETH Zürich	

Conference Activities

The 5th National Algebraic Geometry Conference, University of Peking.	2026
Chicago-Tokyo Algebraic Geometry Workshop, University of Illinois Chicago.	2026
Singularities in Algebraic Geometry Spring School, University of Rennes.	2025
Ontario Graduate Mathematics Conference, University of Waterloo.	2024
Canadian Mathematical Society Summer Meeting, University of Saskatchewan.	2024
Equivariant Methods in Geometry, Cambridge University.	2024
Algebraic Geometry Summer School, Fudan University.	2023
Algebraic Geometry Summer School, Fudan University.	2021

Languages

Chinese: Native speaker.
English: Fluent.
Japanese: Fluent.